Year 10 Foundation

REVISION PACK

Algebra 2

Name: _____

Class: _____

Algebra 2 · Indices in algebra (Lesson 7.1)

Worked example: simplifying with index laws

Simplify

(a) $x^4 \times x^6$ (b) $y^9 \div y^3$ (c) $(3a^4)^2$ (d) $\dfrac{10a^6b^3}{2a^2b}$

(a)	**Step 1**	Add the powers	$4 + 6 = 10$
	Step 2	Answer	x^{10}
(b)	**Step 1**	Subtract the powers	$9 - 3 = 6$
	Step 2	Answer	y^6
(c)	**Step 1**	Number: square it	$3^2 = 9$
	Step 2	Letter: multiply powers	$4 \times 2 = 8$
	Step 3	Answer	$9a^8$
(d)	**Step 1**	Numbers: divide	$10 \div 2 = 5$
	Step 2	Letters: subtract powers	$a^{6-2} = a^4, \quad b^{3-1} = b^2$
	Step 3	Answer	$5a^4b^2$

Practice: simplifying with index laws

1.1 Simplify

(a) $x^2 \times x^7$ (b) $y^{10} \div y^4$ (c) $(5x^2)^2$ (d) $\dfrac{8m^7n^3}{2m^3n}$

1.2 Simplify

(a) $a^5 \times a^3$ (b) $b^7 \div b^2$ (c) $(2a^4)^3$ (d) $\dfrac{15x^5y^4}{3x^2y}$

1.3 Simplify

(a) $p^4 \times p^6$ (b) $q^9 \div q^3$ (c) $(4y^5)^2$ (d) $\dfrac{12p^9q^2}{4p^4q^2}$

1.4 Simplify

(a) $m^3 \times m^8$ (b) $n^{12} \div n^5$ (c) $(3x^2)^3$ (d) $\dfrac{20a^6b^4}{5a^2b^3}$

1.5 Simplify

(a) $x^6 \times x^4$ (b) $y^{11} \div y^2$ (c) $(6b^3)^2$ (d) $\dfrac{18m^5n^3}{9m^4n}$

1.6 Simplify

(a) $t^7 \times t$ (b) $k^{15} \div k^5$ (c) $(2c^5)^4$ (d) $\dfrac{14x^8y^5}{7x^3y^2}$